Answers on
Booklet - 7

Taken from OCR Physics A past papers

Abbreviations, annotations and conventions used in the Mark Scheme	/	=	alternative and acceptable answers for the same marking point
	;	=	separates marking points
	NOT	=	answers which are not worthy of credit
	( )	=	words which are not essential to gain credit
	____	=	(underlining) key words which **must** be used to gain credit
	ecf	=	error carried forward

1.

(a)

v comp decreases going up	B1
as gravitational force / acceleration is against motion or downwards or -9.8 ms^{-2} or is constant	B1
vertical comp is zero at P	B1
h comp no change	B1
as no resistive force/gravitational force has no effect	B1
v comp increases on way down	B1
as gravitational force / acceleration is in same direction as motion or downwards or + 9.8 ms^{-2} or is constant	B1
vertical comp more than horizontal at start but gets smaller than horizontal as it goes up	B1

Penalise confusion of horizontal and vertical twice and then apply ecf

MAX 5

(b)

k.e. decreases on way up / k.e. increases on the way down	B1
k.e. not zero at top of flight / k.e. min at P (top)	B1
p.e. increases on way up / p.e. max at P / decreases on way down	B1
loss of ke =gain in pe at any point on flight / k.e. converted to p.e.	B1
k.e. = p.e halfway up / pe. + k.e. = constant	B1
k.e. the same as starting value at G	B1
p.e. zero at G / p.e. same as starting value at G	B1

MAX 4

(c)

air resistance/drag slows down the ball or reduces vertical or horizontal component or reduces k.e.	B1
height reached is less	B1
range is less	B1
acceleration is no longer constant	B1
acceleration greater than g when going up or greater than – 9.8 going up or less than 9.8 coming down	B1
h comp reduces when going up and coming down	B1
k.e. at G less than k.e. at T / k.e. transfers to thermal	B1
terminal velocity (may be) reached on the way down	B1

MAX 3

2.

a	i	$W_c = 2.4 \times 10^5 \times \sin 4.8$ $= 2.0 \times 10^4$	M1 A0	Allow W x sin 4.8
b		W = F x d W / t = F x v = 12000 x 15 = 180000 ($J\ s^{-1}$)	 C1 A1	
c		P = W / t P = F x v $P = 3.2 \times 10^4 \times 15$ = 480000 (W)	C1 A1	
d		pe / t = 480000 – 180000 = 300000 allow alternative method of determining the vertical height moved in one second and use of mgh e.g. h = 15 sin4.8 = 1.255 W/t = mgh/t = $2.4 \times 10^5 \times 1.255$ = 300000 unit: W or $J\ s^{-1}$	C1 A1 C1 A1 B1	Do not give a mark for PE = mgh on its own. Do not give a mark for W/t or mgh/t on their own Do not give a mark for W/t or mgh/t on their own Allow $kJ\ s^{-1}$ / kW if consistent with numbers used Allow $N\ m\ s^{-1}$ but not $kg\ m^2\ s^{-3}$
e		less Component of weight acts down the slope	M0 A1	 Weight / force of gravity helps the braking (AW) (not gravity helps braking) Some KE is converted to PE and hence less work has to be done by the braking force

3.

(a)	One reading from the graph e.g. 1.0 N causes 7 mm	C1
	Hence 5.0 (N) causes 35 ± 0.5 (mm) (allow one mark for 35 ± 1 (mm)	A1
(b) (i)	Force on each spring is 2.5 (N)	C1
	extension = 17.5 (mm) allow 18 (mm) or reading from graph [allow ecf from (a)]	A1
(ii)	strain energy = area under graph / ½ F x e	C1
	$= 2 \times 0.5 \times 2.5 \times 17.5 \times 10^{-3}$	
	= 0.044 (J) [allow ecf from (b)(i)]	A1
(c)	E = stress / strain	C1
	Stress = force / area and strain = extension / length	C1
	extension = (F x L) / (A x E)	
	$= (5 \times 0.4) / (2 \times 10^{-7} \times 2 \times 10^{11})$	
	$= 5.(0) \times 10^{-5}$ (m)	A1
(d)	strain <u>energy</u> is larger in the spring	B1
	extension is (very much larger) (for the same force) for the spring	B1

4.

(a)	**(i)**	mass x velocity / mv with symbols defined	1	
	(ii)	$0 = m_1v_1 - m_2v_2$ / $m_1v_1 = m_2v_2$/ $0 = m_1v_1 + m_2v_2$	1	
		$v_1/v_2 = m_2/m_1$ or $- m_2/m_1$ *consistent with line above*	1	**[3]**
		max 1 mark for final expression without line 1		
(b)	**(i)**	$E = 1/_2mv^2$ (= 1.2×10^{-14})	1	
		$v = \sqrt{(2 \times 1.2 \times 10^{-14}/6.7 \times 10^{-27})}$	1	
		$v = 1.9 \times 10^6$ (m s^{-1})		
	(ii)	(mv =) $6.7 \times 10^{-27} \times 1.9 \times 10^6$		
		mv = 1.3×10^{-20} (kg m s^{-1})	1	
	(iii)	1.3×10^{-20} = mv = 4.0×10^{-25} v ecf	1	
		or 3.9×10^{-25} v if mass of α-particle is subtracted		
		v = 3.3×10^4 (m s^{-1})	1	
		or $6.7 \times 10^{-27}/4.0 \times 10^{-25}$ = $v/1.9 \times 10^6$	1	
		v = 3.2×10^4 (m s^{-1})	1	**[5]**
(c)	**(i)**	X is about 15 mm from P	1	
	(ii)	XP arrow direction is a straight line through P	1	
	(iii)	N is about 0.3 mm from P; ecf from (b) (iii)	1	**[3]**

5.

(a)	**(i)**	Two vertical arrows of equal length (by eye) and opposite direction in the same vertical line passing through the ball;	1
		weight/gravity/mg/0.49 N and (normal)reaction/string tension/0.49 N	1
	(ii)	gravity; acts on ball and Earth /AW	2
		contact/reaction forces; between ball and strings/racket	2
	(iii)	(resultant force) = ma; = 0.05 x 2 = 0.1 (N)	2
(b)	**(i)**	$v^2 = u^2 + 2gh$/ ½ mv^2 = mgh to give $v^2 = 2gh$;	1
		$v^2 = 2 \times 9.8 \times 0.8$ to give v = 4.0 (m s^{-1}) accept 3.96	1
	(ii)	mv = 0.20 (kg m s^{-1}) accept 0.198	1
	(iii)	2mv = 0.40 (kg m s^{-1}) accept 0.396	1
	(iv)	2mv/t = 8.0 (N) accept 7.92 *ecf b(iii)*	1

6.

stress $= F/A$
or $= F/(\pi d^2/4)$
or $= F/(\pi d^2)$

ratio = 4 (or 4:1)

E is the same for both wires (as same material) [e.g. $E_P = E_Q$]
strain = stress / E
ratio = 4 (or 4:1) [*must be same as* ***(i)***]

7.

kinetic energy $= \frac{1}{2} mv^2$
$= \frac{1}{2} \times 0.150 \times (25)^2$
$= 46.875 = 47$ J

potential energy (= mgh) $= 0.150 \times 9.81 \times 21$
loss = KE – mgh = 46.875 – (30.9)
= 15.97 = 16 J

work done = 16 J
work done = force × distance
$F = 16 / 21 = 0.76$ N

8.

$v^2 = u^2 + 2as$
$= (8.4)^2 + 2 \times 9.81 \times 5$
$= 12.99$ m s^{-1} *(allow 13 to 2 s.f. but not 12.9)*

$t = (v - u) / a$ or $s = ut + \frac{1}{2}at^2$
$= (12.99 - 8.4) / 9.81$ or $5 = 8.4t + \frac{1}{2} \times 9.81t^2$
$t = 0.468$ s

reasonable shape
suitable scale
correctly plotted 1st and last points at (0,8.4) and (0.88 – 0.96,0)
with non-vertical line at 0.47 s

kinetic energy at end is zero so $\Delta KE = \frac{1}{2} mv^2$ or $\Delta KE = \frac{1}{2} mu^2 - \frac{1}{2} mv^2$
$= \frac{1}{2} \times 0.05 \times (8.4)^2$
$= (-)\ 1.8\,J$

final maximum height $= (4.2)^2 / (2 \times 9.8) = (0.9\ (m))$
change in PE $= mgh_2 - mgh_1$
$= 0.05 \times 9.8 \times (0.9 - 5)$
$= (-)\ 2.0\,J$

change is – 3.8 (J)
energy lost to ground (on impact) / energy of deformation of the ball / thermal energy in ball

9.

constant velocity/zero acceleration and therefore no resultant force
no resultant force (and no resultant torque) hence in equilibrium

<u>component of weight</u> $= 450 \times 9.81 \times \sin 12°$ (= *917.8*)
tension $= 650 + 450\,g \sin 12° = (650 + 917.8)$
$= 1600\ (1570)\,N$

work done against frictional force or friction between log and slope
output power greater than the gain in PE / s

10.

$v = u + at$
$= 4.23 + 9.81 \times 1.51$
$= 19.0(4)\ m\,s^{-1}$ (Allow 2 s.f.)
(*Use of –g max 1/2. Use of g = 10 max 1/2. Allow use of 9.8. Allow 19 m s*$^{-1}$)

either $s = ut + \frac{1}{2} at^2$ (or $v^2 = u^2 + 2as$ etc.)
$= 4.23 \times 1.51 + 0.5 \times 9.81 \times (1.51)^2$
$= 17.6\,m$ (*or 17.5 m*)
(*Use of –g here wrong physics (0/2)*)

$F = \Delta P / \Delta t$ *need idea of* <u>*change*</u> *in momentum*
$= [0.0465 \times (18.6 + 19)] / 12.5 \times 10^{-3}$
$= 140$ N
(*Use of – sign max 2/4. Ignore –ve sign in answer*)
Direction: upwards

$h = ½ \times (18.6)^2 / 9.81$
$= 17.6$ m *(2 s.f. –1)*
(*Use of 19 m s^{-1}, 0/2 wrong physics*)

either kinetic energy of the ball is not conserved on impact
or speed before impact is not equal to speed after hence inelastic

11.

$s = ut + ½ at^2$
$15 = 0.5 \times 9.81 \times t^2$
$T = 1.7$ s

if $g = 10$ is used then –1 but only once on paper

vertical component v_v:
$v_v^2 = u^2 + 2as = 0 + 2 \times 9.81 \times 15$ or $v_v = u + at = 9.81 \times 1.7(5)$
$v_v = 17.16$
resultant velocity: $v^2 = (17.16)^2 + (20)^2$
$v = 26$ m s^{-1}

If $u = 20$ is used instead of $u = 0$ then 0/3
Allow the solution using:
initial (potential energy + kinetic energy) = final kinetic energy